

Bachelor Project



**Czech
Technical
University
in Prague**

F3

**Faculty of Electrical Engineering
Department of Radioelectronics**

Design of a platform for 2D LiDaR

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Field of study: Electronics and Communications

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Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 13. November 2025

Abstract

There are currently many advanced LiDAR sensors available for 3D spatial scanning [*insert citation]. However, most of these sensors are too expensive for the average user. The most affordable commercial option is the Unitree L1, which costs \$ 250 [*insert citation]. The aim of this miniproject is to design a platform for a 360° LiDAR sensor, which will be utilized in a subsequent bachelor's thesis to visualize a 3D image of the scanned environment.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

Supervisor: Ing. Pavel Puričer, Ph.D.

Abstrakt

V současnosti existuje spousta pokročilých lidar-senzorů určených pro 3D snímání prostoru [*insert citation*]. Většina z nich je však cenově nedostupná pro běžného uživatele. Nejvíce dostupným komerčním řešením je Unitree L1, který stojí \$ 250 [*insert citation*]. Cílem toho miniprojektu je navrhnout pro 360° lidar-senzor platformu a použít ji v navazující bakalářské práci pro vizualizaci 3D-obrazu naskenovaného prostoru.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 2D LiDaR

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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing



Chapter 1

Introduction

Nowadays, Light Detection and Ranging (LiDAR) sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [insert citation for robotics BAP and Pinliang book]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the most affordable commercial LiDAR sensors which was found during the research for this miniproject cost approximately \$250[insert Unitree citation]. The primary motivation behind this miniproject (and the subsequent bachelor thesis) is to develop a cost-effective platform for a 360° LiDAR sensor that can be utilized for 3D spatial scanning and visualization of the surrounding environment. A subsequent bachelor thesis will then focus on the Digital Signal Processing (DSP) techniques for effective processing of the acquired point cloud data and proper utilization of open-source visualization tools (such as Python's Open3D library[insert citation of a site or a thesis for that library]) to recreate a 3D scene.



Chapter 2

Conclusion